

Database Management Systems and MySQL

Course Roadmap & Learning Overview

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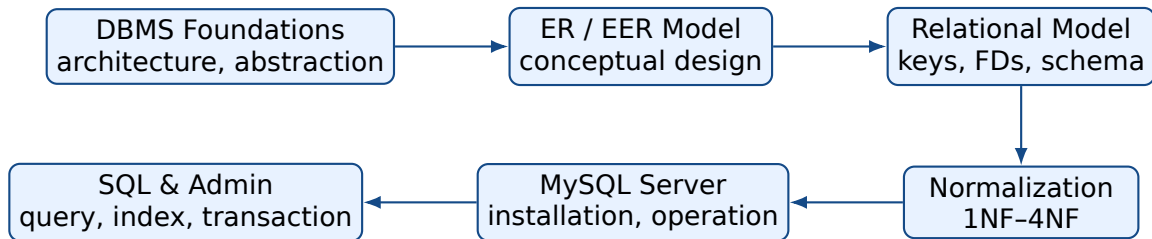
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Course Learning Objectives

By the end of this course, students should be able to:

- Explain the role of **databases** and **DBMSs** in information systems.
- Design conceptual data models using the **ER/EER model**.
- Map ER models to **relational schemas** and analyze keys and functional dependencies.
- Apply **normalization** to reduce redundancy and improve data consistency.
- Use **MySQL** to administer, query, optimize, and program database systems.

Course Big Picture



Main idea: move from data modeling to database implementation and data exploitation with MySQL.

Part 1: Introduction to DBMS

Main topics

- Databases and database management systems.
- Why DBMSs are needed in real-world information systems.
- One-tier, two-tier, and three-tier architectures.
- Data abstraction and data independence.
- Schema, instance, and data models.

Guiding question

Why should a business application avoid storing important data in separate unmanaged files?

What Problems Does a DBMS Solve?

Without a DBMS

- Duplicate data and weak control.
- Difficulty maintaining consistency.
- Manual security and access control.
- Hard backup, recovery, and scaling.

With a DBMS

- Centralized data management.
- Data integrity constraints.
- SQL-based querying.
- Transactions, security, and backup support.

Part 2: Entity Relationship Model

Role

The ER model describes data at the **conceptual** level before it is translated into relational database tables.

Entities

Customers, orders, products.

Attributes

Customer ID, order date, selling price.

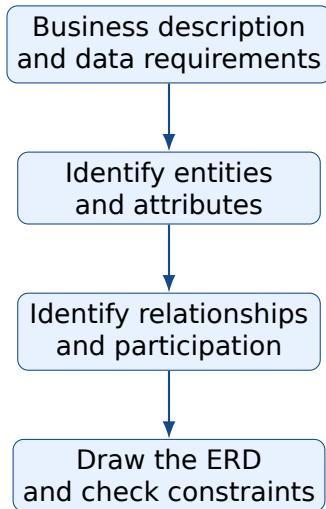
Relationships

A customer places orders; an order contains products.

Practice

Identify entities, attributes, keys, and relationships for a small sales system.

From Business Requirements to ERD



Part 3: Relational Model and Functional Dependencies

Main topics

- Relation, attribute, tuple, and relational schema.
- Super key, candidate key, primary key, and foreign key.
- Mapping from ER model to relational model.
- Functional dependency: determinant and dependent attribute.
- Attribute closure and schema design strategies.

Example

If StudentID uniquely determines StudentName, then:

$$\text{StudentID} \rightarrow \text{StudentName}.$$

Mapping ER Models to Relations

- Each strong entity usually becomes a table.
- The entity key becomes the table primary key.
- A one-to-many relationship is usually represented by a foreign key on the many side.
- A many-to-many relationship is usually converted into an associative table.

Example

```
Customer(CustomerID, Name)
Order(OrderID, OrderDate,
       CustomerID)
```

Design check

A good design preserves business meaning, keys, and relationship constraints.

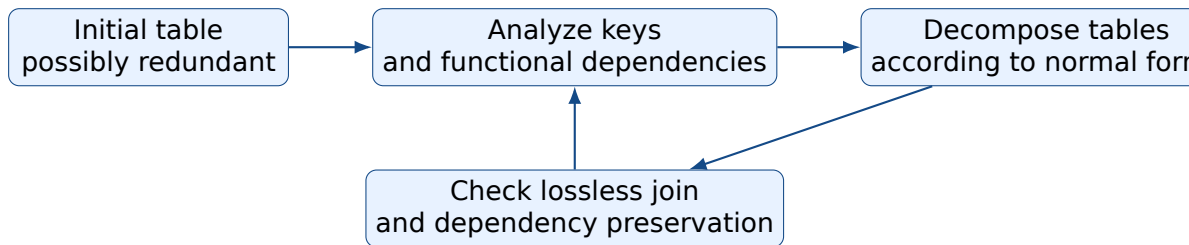
Part 4: Normalization

Goal

Normalization reduces data redundancy and prevents update, insertion, and deletion anomalies.

1NF	Each cell stores an atomic value; repeating groups are removed.
2NF	No partial dependency on part of a composite key.
3NF	No transitive dependency among non-key attributes.
4NF	Independent multivalued dependencies are handled properly.

Normalization Thinking



Part 5: MySQL Server

Implementation topics

- Introduction to MySQL and its tool ecosystem.
- Installing MySQL Server and MySQL Workbench on Windows.
- Connecting through command line and VS Code.
- Starting, stopping, and restarting the MySQL service.
- Loading the `classicmodels` sample database.
- Storage engines, option files, system variables, and the data directory.

Basic MySQL Operations

Create and select a database

```
CREATE DATABASE shop_db  
  CHARACTER SET utf8mb4  
  COLLATE utf8mb4_unicode_ci;
```

```
USE shop_db;  
SHOW TABLES;
```

Safety checklist

Before running DROP DATABASE or DROP TABLE, verify the environment, verify the database/table name, and make sure an appropriate backup exists for important data.

Part 6: SQL Statements in MySQL

Data Definition

- Data types.
- CREATE TABLE.
- Constraints.
- ALTER, RENAME, DROP, TRUNCATE.

Querying Data

- Basic SELECT.
- JOIN, GROUP BY, HAVING.
- Subqueries and CTEs.
- Set operations and window functions.

Sample SELECT Query

Total revenue by customer

```
SELECT c.customerName,  
       SUM(od.quantityOrdered * od.priceEach) AS revenue  
FROM customers c  
JOIN orders o ON c.customerNumber = o.customerNumber  
JOIN orderdetails od ON o.orderNumber = od.orderNumber  
GROUP BY c.customerNumber, c.customerName  
ORDER BY revenue DESC;
```

Indexes, Query Optimization, and Transactions

Index Optimization

- Create, drop, and inspect indexes.
- Composite indexes, unique indexes, prefix indexes, and invisible indexes.
- Read EXPLAIN and use index hints when necessary.

Data Modification & Transactions

- INSERT, UPDATE, DELETE, and advanced data modification.
- START TRANSACTION, COMMIT, ROLLBACK, and SAVEPOINT.
- Locking, row locks, metadata locks, and deadlocks.

Part 7: Views, Procedures, Functions, Triggers, and Events

Core idea

These database objects package processing logic inside the database, reduce repeated application-layer code, and support task automation.

- **View**: packages a query.
- **Stored procedure**: packages a processing workflow.
- **Stored function**: returns a value.
- **Trigger**: runs automatically on INSERT/UPDATE/DELETE.
- **Event**: schedules recurring tasks in MySQL.

Suggested Learning Roadmap

Weeks 1-2: DBMS fundamentals, architecture, abstraction, independence

Weeks 3-4: ER/EER modeling and ERD practice

Weeks 5-6: Relational model, keys, ER-to-relational mapping

Weeks 7-8: Functional dependencies, closure, normalization

Weeks 9-11: MySQL Server, administration, SELECT, JOIN, aggregate

Weeks 12-14: Indexes, transactions, views, procedures, triggers, events

Learning Activities and Assessment

Activities

- Read lecture notes before class.
- Complete MySQL and classicmodels labs.
- Draw ERDs from business scenarios.
- Write queries and explain results.

Outputs

- ERD and relational schema.
- SQL scripts for tables and constraints.
- Analytical SQL query set.
- Short report on design and optimization.

Main References

- Carlos Coronel, Steven Morris. *Database Systems: Design, Implementation, and Management*, 14th Edition. Cengage, 2023.
- Elvis C. Foster, Shripad Godbole. *Database Systems: A Pragmatic Approach*, 3rd Edition. CRC Press / Taylor & Francis, 2022.
- MySQL Tutorial: practical materials on MySQL, SQL, and advanced topics.
- GeeksforGeeks DBMS and SQL tutorials: supplementary topic-based materials.
- Roadmap.sh SQL Roadmap: a self-study roadmap for SQL.

Thank you!

Questions & Discussion